

Diagnosis information - Air-conditioning system

Adjusting motor Faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Faulty
- Current permanent Too high

Fault effects

- ♦ **Adjusting motor** No function

Diagnosis information - Air-conditioning system

Adjusting motor Measurement run

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- **Adjusting motor** Step

too high / too low

Fault effects

- ♦ **Adjusting motor** No function

Diagnosis information - Air-conditioning system

Operating button stuck

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

One **Operating button** for longer than 16 Second(s) pressed

Fault effects

- ♦ Limited function
- ♦ **Operating button** No function

Diagnosis information - Air-conditioning system

Operating button LED Faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

Ignition on

Fault effects

- ♦ Limited function

Diagnosis information - Air-conditioning system

Operating button Faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

Ignition on

Fault effects

- ♦ Limited function

Diagnosis information - Air-conditioning system

Sport/Sportchrono not ready

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- Internal function test

Fault setting condition(s)

Button SPORT / SPORT PLUS pressed

Fault effects

- ♦ Limited function

Diagnosis information - Air-conditioning system

Normal instead of Sport

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- Internal function test

Fault setting condition(s)

Button SPORT / SPORT PLUS pressed

Fault effects

- ♦ Limited function

Diagnosis information - Air-conditioning system

no compressor run-in

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Engine running

Fault setting condition(s)

- compressor run-in not completed
- compressor run-in Active
- no compressor run-in

Fault effects

- ♦ **Air-conditioning system** none Cooling performance

Diagnosis information - Air-conditioning system

Production mode

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Vehicle hand-over or Coding not / incomplete performed

Fault effects

- ♦ **Air-conditioning system** No function

C11005

Diagnosis information - Air-conditioning system

Control unit faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault effects

- ♦ **Air-conditioning system** No function

C11006

Diagnosis information

Air-conditioning system regulator coding

Diagnostic conditions

- Ignition on
- Battery positive voltage OK (11.4 to 14.5 V)
- Internal function test completed on air-conditioning system regulator control unit

Possible fault causes

- No coding of air-conditioning system regulator

Diagnostic information

Air-conditioning system regulator programming

Diagnostic conditions

- Ignition on
- Battery positive voltage OK (11.4 to 14.5 V)
- Internal function test completed on air-conditioning system regulator control unit

Possible fault causes

- ◆ Measurement run for servo motors was not performed or was not completed
- Perform the teaching process for the servo motors in the >PIWIS Tester II<:
>Air conditioning< ⇒ >Maintenance/repairs< ⇒ >Servo motor teaching<

Diagnosis information - Air-conditioning system

Communication **Servo motor** LIN bus 1

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- ◆ no communication LIN bus 1
- ◆ Short circuit to B+/ground/between lines or open circuit in the lines
- ◆ **Servo motor** Faulty

Fault effects

No function of the **Servo motor**

Diagnosis information - Air-conditioning system

Communication **Servo motor** LIN bus 1

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- ◆ no communication LIN bus 1
- ◆ Short circuit to B+/ground/between lines or open circuit in the lines
- ◆ **Servo motor** Faulty

Fault effects

No function of the **Servo motor**

Diagnosis information - Air-conditioning system

Communication **Servo motor** LIN bus 1

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- ◆ no communication LIN bus 1
- ◆ Short circuit to B+/ground/between lines or open circuit in the lines
- ◆ **Servo motor** Faulty

Fault effects

No function of the **Servo motor**

Diagnosis information - Air-conditioning system

Communication **Servo motor** LIN bus 1

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- ◆ no communication LIN bus 1
- ◆ Short circuit to B+/ground/between lines or open circuit in the lines
- ◆ **Servo motor** Faulty

Fault effects

No function of the **Servo motor**

Diagnosis information - Air-conditioning system

Communication **Servo motor** LIN bus 1

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- ◆ no communication LIN bus 1
- ◆ Short circuit to B+/ground/between lines or open circuit in the lines
- ◆ **Servo motor** Faulty

Fault effects

No function of the **Servo motor**

Diagnosis information - Air-conditioning system

Communication **Servo motor** LIN bus 1

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- ◆ no communication LIN bus 1
- ◆ Short circuit to B+/ground/between lines or open circuit in the lines
- ◆ **Servo motor** Faulty

Fault effects

No function of the **Servo motor**

Diagnosis information - Air-conditioning system

Communication **Servo motor** LIN bus 1

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- ♦ no communication LIN bus 1
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ **Servo motor** Faulty

Fault effects

No function of the **Servo motor**

Diagnosis information - Air-conditioning system

Communication **Servo motor** LIN bus 1

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- ◆ no communication LIN bus 1
- ◆ Short circuit to B+/ground/between lines or open circuit in the lines
- ◆ **Servo motor** Faulty

Fault effects

No function of the **Servo motor**

Diagnosis information - Air-conditioning system

Communication with operator control unit LIN bus 2

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- ♦ No communication with operator control unit LIN bus 2
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines

Fault effects

- Operator control unit LIN bus 2 does not function

Diagnosis information - Air-conditioning system

Communication **Air quality sensor** LIN bus 1

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- ♦ no communication LIN bus 1

Time < 4 s

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ **Air quality sensor** Faulty

Fault effects

No function of the **Air quality sensor**

C1300A

Diagnosis information - Air-conditioning system

Communication **Servo motor** LIN bus 1

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

NOTE

- none Communication **Servo motor** , but Communication **Air quality sensor** OK

Fault setting condition(s)

- ♦ no communication Motors LIN bus 1
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines

Fault effects

No function **Servo motor**

C1300B

Diagnosis information - Air-conditioning system

Power supply Fan

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Fan **ON**

Fault setting condition(s)

- Short circuit to B+
- Short circuit to ground
- Open circuit

Fault effects

- ♦ **Air-conditioning system** none Cooling performance

Diagnosis information - Air-conditioning system

Communication with operator control unit LIN bus 2

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- ♦ No communication with operator control unit LIN bus 2
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines

Fault effects

- Operator control unit LIN bus 2 does not function

Diagnosis information - Air-conditioning system

Air quality sensor

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- Internal function test

Fault setting condition(s)

Implausible signal, no communication

- Period of time
 $t < 4 \text{ s}$

Fault effects

- ♦ Limited function

Diagnosis information - Air-conditioning system

Power supply Fan

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Fan **ON**

Fault setting condition(s)

- Short circuit to B+
- Short circuit to ground
- Open circuit

Fault effects

- ♦ **Air-conditioning system** none Cooling performance

C13060

Diagnosis information - Air-conditioning system

LIN bus Data line

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- Internal function test

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line

Fault effects

- ♦ Limited function

C13061

Diagnosis information - Air-conditioning system

LIN bus Data line

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- Internal function test

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line

Fault effects

- ♦ Limited function

Diagnosis information - Air-conditioning system

LIN bus Data line

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- Internal function test

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line

Fault effects

- ♦ Limited function

Diagnosis information - Air-conditioning system

LIN bus Data line

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- Internal function test

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line

Fault effects

- ♦ Limited function

Diagnosis information - Air-conditioning system

Communication **Air quality sensor** LIN bus 1

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- ♦ no communication LIN bus 1

Time < 4 s

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ **Air quality sensor** Faulty

Fault effects

No function of the **Air quality sensor**

C13066

Diagnosis information - Air-conditioning system

Operating button Faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

Ignition on

Fault effects

- ♦ Limited function

Diagnosis information - Air-conditioning system

Communication **Air quality sensor** LIN bus 1

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- ♦ no communication LIN bus 1

Time < 4 s

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ **Air quality sensor** Faulty

Fault effects

No function of the **Air quality sensor**

Diagnosis information - Air-conditioning system

Communication with operator control unit LIN bus 2

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- ♦ No communication with operator control unit LIN bus 2
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines

Fault effects

- Operator control unit LIN bus 2 does not function

C1306A

Diagnosis information - Air-conditioning system

Power supply Fan

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Fan **ON**

Fault setting condition(s)

- Short circuit to B+
- Short circuit to ground
- Open circuit

Fault effects

- ♦ **Air-conditioning system** none Cooling performance

C1306C

Diagnosis information - Air-conditioning system

Power supply Fan

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Fan **ON**

Fault setting condition(s)

- Short circuit to B+
- Short circuit to ground
- Open circuit

Fault effects

- ♦ **Air-conditioning system** none Cooling performance